

# Fehlerrechnung

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**Gegeben:**  $a = 15 \pm 2$ ,  $b = 20 \pm 3$ ,  $c = 10 \pm 2$

$$f = a + 2(b + c) = a + 2b + 2c$$

$$f = 15 + 2(20 + 10) = 15 + 60 = 75$$

$$\Delta f = \sqrt{(\Delta a)^2 + (2\Delta b)^2 + (2\Delta c)^2}$$

$$\Delta f = \sqrt{2^2 + (2 \cdot 3)^2 + (2 \cdot 2)^2} = \sqrt{4 + 36 + 16} = \sqrt{56} \approx 7,48$$

$$\frac{\Delta f}{f} = \frac{7,48}{75} \approx 0,10 = 10\%$$

$$f = 75 \pm 7,5 \quad (\text{relativer Fehler} \approx 10\%)$$

**Gegeben:**  $a = 15 \pm 2$ ,  $b = 20 \pm 3$ ,  $c = 10 \pm 2$

$$g = \frac{b^2}{a \times c}$$

$$g = \frac{20^2}{15 \times 10} = \frac{400}{150} = 2,67$$

$$\left(\frac{\Delta g}{g}\right)^2 = \left(2 \cdot \frac{\Delta b}{b}\right)^2 + \left(\frac{\Delta a}{a}\right)^2 + \left(\frac{\Delta c}{c}\right)^2$$

$$\frac{\Delta b}{b} = \frac{3}{20} = 0,15, \quad \frac{\Delta a}{a} = \frac{2}{15} = 0,1333, \quad \frac{\Delta c}{c} = \frac{2}{10} = 0,2$$

$$\left(\frac{\Delta g}{g}\right)^2 = (2 \cdot 0,15)^2 + 0,1333^2 + 0,2^2 = 0,09 + 0,0178 + 0,04 = 0,1478$$

$$\frac{\Delta g}{g} = \sqrt{0,1478} \approx 0,384$$

$$\Delta g = g \cdot \frac{\Delta g}{g} = 2,67 \times 0,384 = 1,03$$

$$g = 2,67 \pm 1,03 \quad (\text{relativer Fehler} \approx 38\%)$$

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**Gegeben:**  $y' = \frac{2y}{x}, \quad y(1) = 2$

$$\frac{dy}{dx} = \frac{2y}{x}$$

$$\frac{dx}{dy} = \frac{x}{2}$$

$$\int \frac{dy}{y} = \int \frac{2}{x} dx$$

$$\ln |y| = 2 \ln |x| + \tilde{c}$$

$$y = c \times x^2$$

$$y(1) = 2 \rightarrow 2 = c \times x^2$$

$$\boxed{y = 2x^2}$$

**Gegeben:**  $y' \times x = 1 + 2y, \quad y(0) = 5$

$$\Rightarrow \frac{dy}{dx} = \frac{1 + 2y}{x}$$

$$\frac{dy}{dx} - \frac{2}{x}y = \frac{1}{x}$$

$$x^{-2} \frac{dy}{dx} - \frac{2}{x^3}y = \frac{1}{x^3}$$

$$\frac{d}{dx}(y x^{-2}) = \frac{1}{x^3}$$

$$\int \frac{d}{dx}(y x^{-2}) dx = \int \frac{1}{x^3} dx$$

$$y x^{-2} = \int x^{-3} dx = -\frac{1}{2x^2} + C$$

$$y = -\frac{1}{2} + Cx^2$$

Anfangsbedingung:  $y(0) = 5$

$$\boxed{y = -\frac{1}{2} + Cx^2, \quad x \neq 0}$$